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Bài tập khai thác mẫu

Đề tài: Khai thác dữ liệu và ứng dụng

Môn học: Khai thác dữ liệu và ứng dụng

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Câu 4.1

Suppose you have the set $\mathcal{C}$ of all frequent closed itemsets on a data set $\mathcal{D}$, as well as the support count for each frequent closed itemset. Describe an algorithm to determine whether a given itemset $\mathcal{X}$ is frequent or not, and the support of $\mathcal{X}$ if it is frequent.

Lời giải

Câu 4.2

An itemset $\mathcal{X}$ is called a generator on a data set $\mathcal{D}$ if there does not exist a proper subitemset $\mathcal{Y} \subset \mathcal{X}$ such that $\text{sup}(\mathcal{X}) = \text{sup}(\mathcal{Y})$. A generator $\mathcal{X}$ is a frequent generator if $\text{sup}(\mathcal{X})$ passes the minimum support threshold. Let $\mathcal{G}$ be the set of all frequent generators on a data set $\mathcal{D}$.

- Can you determine whether an itemset $\mathcal{A}$ is frequent and the support of $\mathcal{A}$, if it is frequent, using only $\mathcal{G}$ and the support counts of all frequent generators? If yes, present your algorithm. Otherwise, what other information is needed? Can you give an algorithm assuming the information needed is available?
- What is the relationship between closed itemsets and generators?

Lời giải

Câu 4.3

The Apriori algorithm makes use of prior knowledge of subset support properties.

- Prove that all nonempty subsets of a frequent itemset must also be frequent.
- Prove that the support of any nonempty subset s' of itemset s must be at least as great as the support of s .
- Given frequent itemset l and subset s of l , prove that the confidence of the rule " $s' \Rightarrow (l - s')$ " cannot be more than the confidence of " $s \Rightarrow (l - s)$ ", where s' is a subset of s .

- d. A partitioning variation of Apriori subdivides the transactions of a database $\mathcal{D}$ into n nonoverlapping partitions. Prove that any itemset that is frequent in $\mathcal{D}$ must be frequent in at least one partition of $\mathcal{D}$.

Lời giải

Câu 4.4

Let c be a candidate itemset in C_k generated by the Apriori algorithm. How many length- $(k - 1)$ subsets do we need to check in the prune step? Per your previous answer, can you give an improved version of procedure `has_infrequent_subset` in Fig. 4.4?

Lời giải

Câu 4.5

Section 4.2.2 describes a method for generating association rules from frequent itemsets. Propose a more efficient method. Explain why it is more efficient than the one proposed there. (Hint: consider incorporating the properties of Exercises 4.3(b), (c) into your design.)

Lời giải

Câu 4.6

A database has five transactions. Let `min_sup` = 60% and `min_conf` = 80%.

TID	items_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, K, I, E}

- Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.
- List all the strong association rules (with support s and confidence c) matching the following metarule, where X is a variable representing customers, and $item_i$ denotes variables representing items:

$$\begin{aligned} \forall x \in transaction, buys(X, item_1) \wedge buys(X, item_2) \\ \Rightarrow buys(X, item_3) \quad [s, c] \end{aligned}$$

Lời giải

Câu 4.8

A database has four transactions. Let $\min_sup = 60\%$ and $\min_conf = 80\%$.

cust_ID	TID	items_bought (in the form of brand-item_category)
01	T100	{King's-Crab, Sunset-Milk, Dairyland-Cheese, Best-Bread}
02	T200	{Best-Cheese, Dairyland-Milk, Goldenfarm-Apple, Tasty-Pie, Wonder-Bread}
01	T300	{Westcoast-Apple, Dairyland-Milk, Wonder-Bread, Tasty-Pie}
03	T400	{Wonder-Bread, Sunset-Milk, Dairyland-Cheese}

- At the granularity of item_category (e.g., $item_i$ could be “Milk”), for the rule template,

$$\begin{aligned} \forall X \in transaction, buys(X, item_1) \wedge buys(X, item_2) \\ \Rightarrow buys(X, item_3) \quad [s, c], \end{aligned}$$

list the frequent k -itemset for the largest k , and all the strong association rules (with their support s and confidence c) containing the frequent k -itemset for the largest k .

- At the granularity of brand-item_category (e.g., $item_i$ could be “Sunset-Milk”), for the rule template,

$$\forall X \in customer, buys(X, item_1) \wedge buys(X, item_2) \Rightarrow buys(X, item_3),$$

list the frequent k -itemset for the largest k (but do not print any rules).

Lời giải

Câu 4.9

Suppose that a large store has a transactional database that is distributed among four locations. Transactions in each component database have the same format, namely $T_j : \{i_1, \dots, i_m\}$, where T_j is a transaction identifier, and i_k ($1 \leq k \leq m$) is the identifier of an item purchased in the transaction. Propose an efficient algorithm to mine global association rules. Your algorithm should not require shipping all the data to one site and should not cause excessive network communication overhead.

Lời giải

Câu 4.10

Suppose that frequent itemsets are saved for a large transactional database, DB . Discuss how to efficiently mine the (global) association rules under the same minimum support threshold, if a set of new transactions, denoted as ΔDB , is (incrementally) added in.

Lời giải

Câu 4.11

Most frequent pattern mining algorithms consider only distinct items in a transaction. However, multiple occurrences of an item in the same shopping basket, such as four cakes and three jugs of milk, can be important in transactional data analysis. How can one mine frequent itemsets efficiently considering multiple occurrences of items? Propose modifications to the well-known algorithms, such as Apriori and FP-growth, to adapt to such a situation.

Lời giải

Câu 4.13

Give a short example to show that items in a strong association rule actually may be negatively correlated.

Lời giải

Câu 4.14

The following contingency table summarizes supermarket transaction data, where *hot dogs* refers to the transactions containing hot dogs, $\overline{\text{hot dogs}}$ refers to the transactions that do not contain hot dogs, *hamburgers* refers to the transactions containing hamburgers, and $\overline{\text{hamburgers}}$ refers to the transactions that do not contain hamburgers.

	<i>hot dogs</i>	$\overline{\text{hot dogs}}$	Σ_{row}
<i>hamburgers</i>	2000	500	2500
$\overline{\text{hamburgers}}$	1000	1500	2500
Σ_{col}	3000	2000	5000

- Suppose that the association rule “*hot dogs* $\Rightarrow$ *hamburgers*” is mined. Given a minimum support threshold of 25% and a minimum confidence threshold of 50%, is this association rule strong?
- Based on the given data, is the purchase of hot dogs independent of the purchase of hamburgers? If not, what kind of correlation relationship exists between the two?
- Compare the use of the all_confidence, max_confidence, Kulczynski, and cosine measures with lift and correlation on the given data.

Lời giải